MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 28 – 30 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space and count outcomes to compute probabilities for equally likely outcomes, and apply the complement rule. (CLO 3) **2.** Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard
3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ, σ, p, ρ vs. $\bar{x}$, s, $\hat{p}$, r). (CLO 5) **4.** Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CLO 5) **5.** Explain the standard error as the variability of a statistic and predict how sample size affects it. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	Online Homework: 3.1 Sampling Distributions (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 3 (due Mon Oct 5, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)		Participation: Regression Line – Mon: used-car price problem wrapping up 2.5 (Week 5 objectives)	Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables.